QUANT RESEARCH CASE STUDY | 2026

Market Volatility x Factor Robustness

A walk-forward test of whether volatility regimes change factor payoffs - and whether a simple ex-ante risk budget reduces drawdown in a diversified factor sleeve after costs.

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Executive finding

Factor behavior is state-dependent, but not every apparent regime gap is statistically robust. The largest Sharpe deterioration from low to high volatility occurs in **RMW**. A pre-specified de-risking rule lowers annualized volatility from **4.6%** to **2.8%** and changes maximum drawdown from **-12.8%** to **-6.3%**, after a 2 bp turnover cost. The cost is lower annualized return and Sharpe; the result supports a risk-control use case, not an alpha claim.

OBSERVATIONS	FACTORS	ROBUST HIGH-LOW GAPS
3,895 days	6	2 / 6

Observation -> hypothesis

Market stress does more than raise total volatility. Funding constraints, deleveraging, crowded positioning and shifts in investor risk tolerance can alter the return distribution of long-short equity factors. **H1:** conditional factor Sharpe ratios differ across volatility states. **H2:** reducing aggregate factor exposure in high-volatility states can improve drawdown control without using same-day information.

Decision relevance: the project asks whether a simple regime signal is useful for portfolio risk budgeting, while explicitly rejecting any claim that conditional averages alone prove a tradable edge.

1. Methodology and anti-leakage rules

- **Inputs.** Daily Fama-French five factors plus momentum and the official Cboe VIX close.
- **Ex-ante state.** Day t uses VIX observed on t-1. Low/Mid/High cutoffs are rolling 33rd/67th percentiles over the preceding 756 trading days, with a 252-day warm-up.
- **Descriptive test.** Compare annualized return, volatility and Sharpe by volatility state.
- **Uncertainty.** Estimate 95% intervals for High-minus-Low annualized mean returns with 2,000 moving-block bootstrap draws using 20-day blocks.
- **Execution rule.** Equal-weight six-factor sleeve; target 5% ex-ante volatility from trailing 63-day data; risk budgets 1.00/0.75/0.50 in Low/Mid/High states; no leverage above 1.00; 2 bp per unit of exposure turnover.
- **Validation.** Full sample, 2010-2019, 2020-latest and VIX>=30 stress days are reported separately.

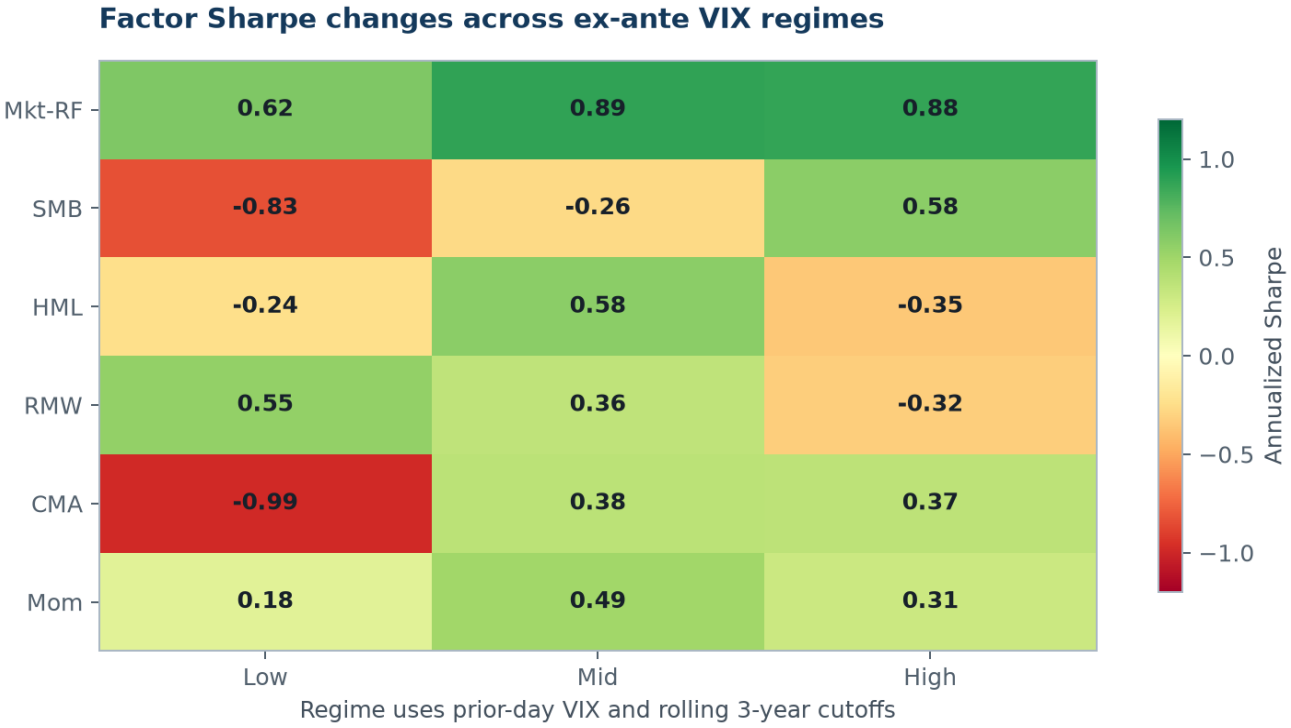
Research design

Input	Transformation	Output / test
FF factors + momentum	Daily decimals; aligned on date	Regime-conditional factor metrics
Cboe VIX close	Lag 1 day; rolling 756-day thresholds	Low / Mid / High state known ex ante
Factor sleeve	63-day volatility target + state budget	Net return after turnover cost
Daily dependence	20-day moving blocks, 2,000 draws	95% CI for High-minus-Low means

Reproducibility controls

- Bootstrap seed: 20260824.
- All thresholds and risk estimates are calculated from lagged or prior-window data.
- Raw inputs, processed daily panel, output tables, charts and source code are retained together.
- The reported trade-off is not optimized away: lower drawdown is accompanied by lower return and Sharpe.

2. Factor behavior by volatility state



Each day's state is known before that day's factor return. Conditional Sharpe varies materially, but the pattern is not monotonic across every factor and should not be interpreted as causal evidence.

State	Factor	N	Ann. return	Ann. vol	Sharpe
Low	Mkt-RF	1498	6.5%	10.5%	0.62
Low	SMB	1498	-7.5%	9.0%	-0.83
Low	HML	1498	-2.0%	8.4%	-0.24
Low	RMW	1498	3.3%	6.0%	0.55
Low	CMA	1498	-5.5%	5.6%	-0.99
Low	Mom	1498	1.9%	10.7%	0.18
Mid	Mkt-RF	1215	12.0%	13.6%	0.89
Mid	SMB	1215	-2.5%	9.6%	-0.26
Mid	HML	1215	7.0%	12.0%	0.58
Mid	RMW	1215	3.1%	8.5%	0.36
Mid	CMA	1215	2.7%	7.2%	0.38
Mid	Mom	1215	7.2%	14.7%	0.49
High	Mkt-RF	1182	23.1%	26.4%	0.88
High	SMB	1182	6.9%	11.9%	0.58
High	HML	1182	-5.9%	16.7%	-0.35
High	RMW	1182	-3.1%	9.6%	-0.32
High	CMA	1182	3.0%	8.1%	0.37
High	Mom	1182	6.6%	21.7%	0.31

3. Robustness: confidence intervals matter

Moving-block bootstrap resampling preserves short-run dependence better than independent daily resampling. Only intervals excluding zero are treated as robust in this diagnostic.

Factor	High - Low return	95% CI low	95% CI high	Excludes zero
Mkt-RF	16.6%	-5.7%	38.6%	No
SMB	14.4%	2.1%	26.9%	Yes
HML	-3.9%	-19.6%	10.7%	No
RMW	-6.4%	-14.5%	2.6%	No
CMA	8.5%	0.3%	16.7%	Yes
Mom	4.7%	-15.0%	23.6%	No

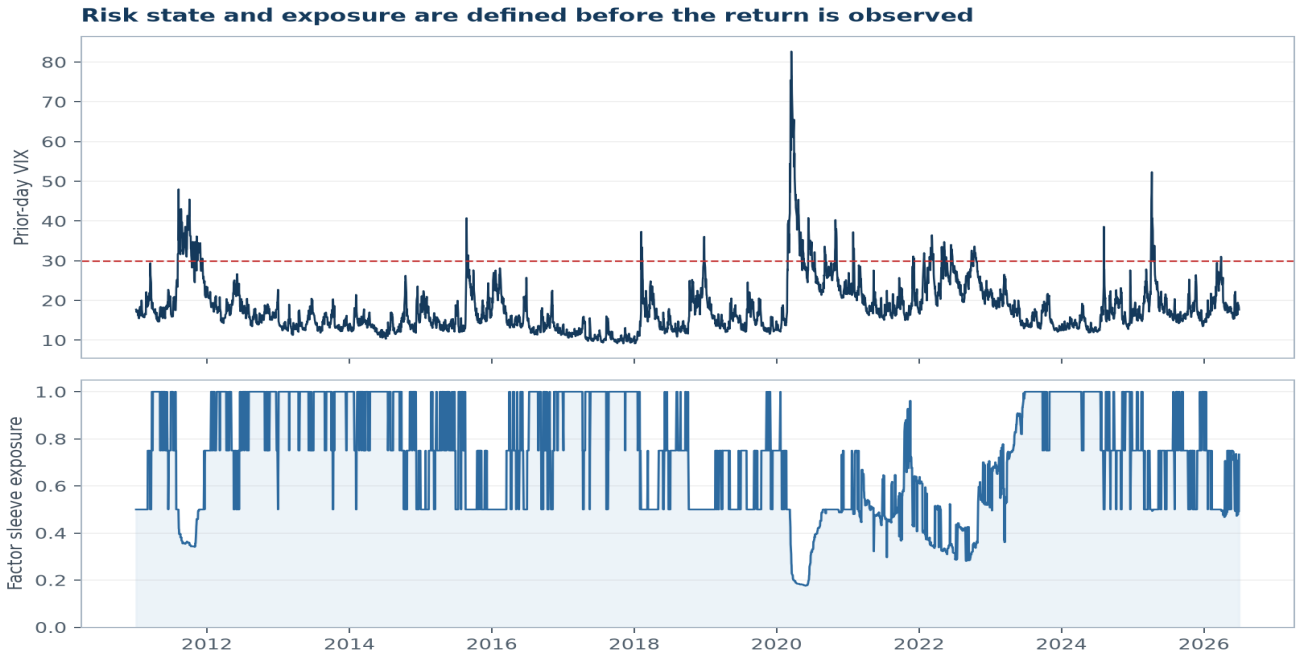
Interpretation

Robust High-minus-Low mean-return gaps appear in **SMB, CMA**. RMW shows the largest deterioration in conditional Sharpe, but its mean-return interval still crosses zero. That distinction prevents the analysis from turning a noisy visual pattern into an overstated signal.

What this validates

- Volatility state is a useful conditioning variable for risk diagnostics.
- A factor-specific regime claim needs uncertainty intervals, not only point estimates.
- The hypothesis is only partially supported; the evidence does not justify tactical factor timing by itself.

4. Walk-forward execution framework



Exposure is a deterministic function of lagged VIX, rolling thresholds, trailing sleeve volatility and the pre-specified state budget. Same-day information is never used.



Both series represent long-short factor sleeves. They are research diagnostics and omit the full implementation constraints of trading individual securities.

5. Validation, decision and limitations

Window	Strategy	N	Ann. return	Ann. vol	Sharpe	Max DD
Full sample	Static EW	3895	2.9%	4.6%	0.62	-12.8%
Full sample	Regime aware	3895	1.4%	2.8%	0.50	-6.3%
2010-2019	Static EW	2264	2.3%	3.1%	0.72	-4.6%
2010-2019	Regime aware	2264	1.3%	2.4%	0.54	-3.1%
2020-latest	Static EW	1631	3.7%	6.1%	0.61	-11.0%
2020-latest	Regime aware	1631	1.6%	3.3%	0.48	-5.8%
VIX >= 30	Static EW	233	3.5%	9.4%	0.37	-8.3%
VIX >= 30	Regime aware	233	0.2%	3.3%	0.08	-3.2%

Decision

The regime-aware sleeve is useful as a transparent drawdown-control overlay, not as a stand-alone return-enhancement strategy. The full-sample maximum drawdown improves from -12.8% to -6.3%, but annualized return falls from 2.9% to 1.4% and Sharpe falls from 0.62 to 0.50. A real portfolio owner should adopt the rule only if drawdown reduction is worth the lower capital utilization.

Limitations and next iteration

- Fama-French factors are research series, not directly investable portfolios; actual financing, borrow, capacity and tax constraints are omitted.
- VIX is a U.S. equity volatility proxy; the conclusion should not be transported to China A-shares without local volatility and factor data.
- The 5% target and 1.00/0.75/0.50 budgets are pre-specified rather than optimized, which reduces overfitting but does not prove optimality.
- Next validation: rolling out-of-sample parameter sensitivity, alternative volatility measures, multiple-testing control and security-level implementation costs.

Reproducible package

The package contains source code, official raw inputs, a processed daily panel, output tables, charts and run metadata. Run `python src/analyze.py` from the project directory after installing the pinned requirements. Portfolio home: https://zengjingrou.netlify.app